

## LECTURE 24 (DECEMBER 9TH)

**Recall.** For  $\operatorname{Re}(s) > 1$ ,

$$\xi(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

converges and for  $\operatorname{Re}(s) > 0$ ,

$$\eta(s) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n^s}$$

converges. If  $\operatorname{Re}(s) > 1$ ,

$$\xi(s) = (1 - 2^{1-s})^{-1} \eta(s)$$

which is a analytic continuation of  $\xi$  to  $\operatorname{Re}(s) > 0$ .

**Proposition.** (Riemann hypothesis) All zeros of such  $\xi$  has the form

$$z = \frac{1}{2} + it$$

**Theorem.** (Euler's theorem) Let  $\xi(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$  for  $s > 1$ ,  $s \in \mathbf{R}$ .

$$\frac{1}{\xi(s)} = \prod_p \left(1 - \frac{1}{p^s}\right)$$

*Proof.*

$$\xi(s) = 1 + \frac{1}{2^s} + \frac{1}{3^s} + \dots$$

Subtracting

$$\xi(s) \frac{1}{2^s} = \frac{1}{2^s} + \frac{1}{4^s} + \frac{1}{6^s} + \dots$$

we obtain

$$\xi(s) \left(1 - \frac{1}{2^s}\right) = 1 + \frac{1}{3^s} + \frac{1}{5^s} + \dots$$

and

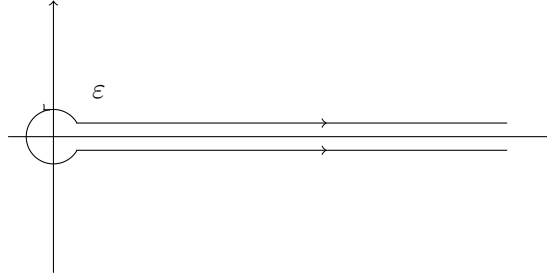
$$\xi(s) \left(1 - \frac{1}{2^s}\right) \left(1 - \frac{1}{3^s}\right) = 1 + \frac{1}{5^s} + \frac{1}{7^s} + \dots$$

Thus,

$$\xi(s) \left(1 - \frac{1}{2^s}\right) \left(1 - \frac{1}{3^s}\right) \dots \left(1 - \frac{1}{p^s}\right) = 1$$

□

**Definition.** (Hankel contour) The Hankel contour is defined as



for  $\varepsilon \rightarrow 0$ .

**Example.** Recall our result for  $0 < \alpha < 1$ ,

$$\int_0^\infty \frac{x^{\alpha-1}}{1+x} dx = \frac{\pi}{\sin \alpha \pi}$$

Notice that

$$f(z) = \frac{z^{\alpha-1}}{1+z}$$

is analytic inside on  $C_{\varepsilon,R}$ , and vanishes on the outer rim. Inside the small key,

$$\left| \int_{\text{IV}} f(z) dz \right| \leq \int_0^{2\pi} \frac{\varepsilon^{\alpha-1}}{1-\varepsilon} \varepsilon dt = 2\pi \frac{\varepsilon^\alpha}{1-\varepsilon} \rightarrow 0$$

as  $\varepsilon \rightarrow 0$ . Now

$$(1 - e^{2\pi\alpha i}) \int_0^\infty \frac{x^{\alpha-1}}{1+x} dx = 2\pi i \operatorname{Res}_{z=-1} f(z)$$

Meanwhile,

$$\operatorname{Res}_{z=-1} (e^{\pi i})^{\alpha-1} = -e^{\alpha\pi i}$$

and through algebra,

$$\int_0^\infty \frac{x^{\alpha-1}}{1+x} dx = 2\pi i \frac{e^{\alpha\pi i}}{e^{2\pi\alpha i} - 1} = \frac{\pi}{\sin \alpha \pi}$$

**Recall.** We have previously seen that for  $\operatorname{Re}(s) > 1$ ,

$$\xi(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{x^{s-1}}{e^x - 1} ds$$

For  $\operatorname{Re}(s) > 1$ , take

$$f(z) = \frac{z^{s-1}}{e^z - 1}$$

Now,

$$\int_H \frac{z^{s-1}}{e^z - 1} dz = (e^{2\pi i s} - 1) \int_0^\infty \frac{x^{s-1}}{e^x - 1} dx$$

A critical point is that the left function is entire, and that the right integral becomes  $\Gamma(s)\xi(s)$ . We now see that the left integral can be used to analytically continue the

right. We now see that the left integral can be used to analytically continue the right. When  $\text{Re}(s) < 0$ ,

$$\begin{aligned} \int_H \frac{z^{s-1}}{e^z - 1} dz &= \lim_{R \rightarrow \infty} \int_{C_{\epsilon, R}} \frac{z^{s-1}}{e^z - 1} dz = -2\pi i \sum_{z=2n\pi i} \text{Res} f(z) \\ &= -2\pi i \sum_{n=-\infty}^{\infty} (2n\pi i)^{s-1} = -2\pi i (2\pi i)^{s-1} \sum_{n=1}^{\infty} \left[ \frac{1}{n^{1-s}} + (-1)^{s-1} \frac{1}{n^{1-s}} \right] \\ &= -(2\pi i)^s \left[ \xi(1-s) + (-1)^{s-1} \xi(1-s) \right] \end{aligned}$$

and we close  $H$  by adding a big circle with radius  $\infty$ . Then, we can obtain the values using the residue theorem with  $n \in \mathbf{Z} \setminus \{0\}$ . Notice how the right converges when  $\text{Re}(s) < 0$ .

**Corollary.** Therefore,

$$\xi(s) = 2^s \pi^{s-1} \sin \frac{\pi s}{2} \xi(1-s) \Gamma(1-s)$$

determines all of  $\xi(s)$  and has a simple pole at  $s = 1$  with residue 1, otherwise analytic. Notice that

$$\xi(s) = 0$$

if  $s = -2, -4, -6, \dots$ . Also, if  $\xi(\frac{1}{2} + it) = 0$ ,  $\xi(\frac{1}{2} - it) = 0$ . The Riemann hypothesis in this form is that  $\xi(s) = 0$  has all solutions on  $0 < \text{Re}(s) < 1$  of the form

$$s = \frac{1}{2} + it$$

where the band is called the critical strip. The above is called the critical line. The fact that  $\xi(s) = 0$  has no zeros on the line

$$s = 1 + it$$

proves the prime number theorem,

$$\lim_{n \rightarrow \infty} \frac{\pi(n) \log n}{n} = 1$$